

What Is a Computer? — Detailed Lecture Notes

JKSSB Computer Awareness

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Video lesson: [What is a Computer?](#)

1.1 A computer is a data-processing system

Start with an office task. A clerk enters marks into a spreadsheet, a formula adds the values and calculates an average, the result appears on screen, and the workbook can be saved. A computer is not merely the screen or the box on the desk. The task depends on data, instructions, processing, an output path, and possibly storage.

Definition (Computer). *An electronic, programmable system that accepts data, processes it under instructions, produces information, and can store data or results.*

Each part of the definition matters. *Electronic* describes the technology used by the system. *Programmable* means that instructions specify the operations to perform. The system accepts inputs, applies those instructions, and makes results available. It can also retain data or results for later use.

Definition (Data). *Raw facts or symbols supplied to a system before they have been organized for the particular question being asked.*

Definition (Information). *Data that have been organized or processed so that they are useful for a particular purpose.*

Definition (Instruction). *A rule that tells the computer what operation to perform or what action to take.*

Definition (Program). *An organized set of instructions that directs a computer through a task.*

Data and information are defined by their role in a task, not by whether the values are numbers or text. Individual marks are data when the question is “What are the three scores?” Their average is information when the question is “What was the student’s average?” The average can itself become input data for a later task, so the same value may play different roles in different contexts.

Example 1.1.1. *A clerk enters 68, 74, and 81. These three marks are the input data. Adding them gives 223; dividing by three gives $223/3 = 74.333\dots$, or approximately 74.33 to two decimal places. The total and average are processed results. They are useful information if the task is to summarize performance. Saving the workbook retains the records.*

Example 1.1.2. *Suppose the same three marks are each out of 100. The total possible score is therefore 300. The percentage is*

$$\frac{68 + 74 + 81}{300} \times 100 = \frac{223}{300} \times 100 = 74.333\dots\%$$

Rounded to two decimal places, the percentage is 74.33%. The total, maximum possible score, formula, and rounding rule all matter; the computer cannot infer a missing maximum score from the marks alone.

1.2 Input, processing, output, and storage

Definition (Input). *Data and instructions that enter a computer system through a device, file, network, or software interface.*

Definition (Processing). *Operations carried out on input data according to instructions, usually by the processor and program involved in the task.*

Definition (Output). *The result made available by a computer system, for example on a display, in a printed document, as sound, or through another system.*

Definition (Storage). *The retention of data, instructions, or results for current or later use.*

The basic three-stage model is IPO: Input, Processing, Output. Many introductory diagrams add Storage, giving IPOS. Storage is a distinct function, not another name for processing. The stages describe roles in a task; they do not require every computer job to follow a single rigid, one-direction path. A saved result can be read back as input to a later operation.

Stage	Question to ask	Marks-spreadsheet example
Input	What data or instructions enter?	The three marks and the average formula
Processing	What operation changes or organizes the data?	Addition, division, and percentage calculation
Output	What result is presented, and how?	The total or percentage shown on screen or printed
Storage	What is retained for later use?	The saved workbook and its records

The physical item does not determine the stage by itself. A monitor is normally an output device, but the classification question concerns the monitor's role in the described task. A saved file is stored data; opening that same file later makes its contents input to the new task. Likewise, information describes the usefulness of processed data, whereas output describes how the result is made available. A screen can present useful information, but the terms answer different questions.

Example 1.2.1. *In the marks spreadsheet, typing or importing the scores is input. The formula performs processing. The displayed total and average are output. Saving the workbook is storage. If the saved workbook is opened tomorrow, reading it is input to that later session.*

1.3 Characteristics and limitations

Definition (Speed). *The ability to carry out many operations in a short time.*

Definition (Accuracy). *The ability to produce consistent, precise results when the data, instructions, and operation are correct.*

Definition (Diligence). *The ability to repeat work without tiring in the human sense, provided the system continues to receive power and function correctly.*

Definition (Versatility). *The ability to perform different kinds of tasks by running different programs.*

Definition (Automation). *The ability to continue a programmed operation after it has been started and the required inputs or conditions have been supplied.*

Definition (Storage as a characteristic). *The ability to retain data, instructions, and results for later access.*

These characteristics describe capabilities, not a guarantee that every result is correct or useful. Speed concerns how quickly operations are performed; accuracy concerns whether the operation is correctly carried out on the supplied data. Storage concerns retaining material; it does not validate that material. Versatility comes from using different programs, while automation means a started procedure can continue according to its instructions and conditions.

Speed and accuracy must not be confused with judgement. A computer can calculate quickly and repeat the same operation consistently, but it cannot infer that a deliberately or accidentally incorrect mark is false merely because the value is stored in a spreadsheet. Accuracy is conditional on the quality of data and instructions.

Definition (GIGO). *“Garbage In, Garbage Out”: unsuitable or incorrect input can lead to unsuitable or incorrect output, even when the computer follows its instructions.*

Example 1.3.1. *Suppose the correct marks are 68, 74, and 81, but a user enters 18 instead of 81. The formula may correctly compute $68 + 74 + 18 = 160$ and $160/3 \approx 53.33$. The arithmetic is consistent with the entered values, but the output is not a valid summary of the actual marks. The error entered at the input stage was not corrected by processing.*

The example illustrates the limitation often summarized by GIGO. If the input is wrong, a correct formula can still produce an incorrect result. If the input is correct but the formula is wrong, the result can also be wrong. The machine may execute either set of instructions consistently; it does not automatically decide that the source data are truthful, complete, or fair. Computers do not possess human judgement or common-sense understanding by default, so a person must interpret output in context and decide what action it justifies.

1.4 Applications and role classification

The same IPO/IPOS model applies across fields. What changes is the type of data and the useful result; the roles of input, processing, output, and storage remain identifiable.

Field	Input	Processing / output	Storage
Office	Text, figures, and instructions	Create a document or calculate a total	Document or work-book file
Banking	Account and transaction details	Validate and process a transaction; show a status	Account and transaction records
Education	Learner responses and assessment rules	Score work and present results or learning resources	Assessment and learner records
Healthcare	Patient details and appointment requests	Organize records, schedules, or diagnostic support	Patient and appointment records
Governance	Online form fields and supporting details	Process a request and provide a citizen service	Submitted forms and service records

For exam questions, classify the role within the described task. A keyboard is an input device; the formula is processing; a monitor presents output; a saved file is stored data. A physical object

may support more than one stage, but the question usually asks for the role it is playing in that particular scenario.

Example 1.4.1. *An online expense form accepts entered amounts, checks and sums them using the programmed rules, displays the total, and stores the submitted record. If one amount is mistyped, the calculation may still be executed correctly on the wrong data. The user should correct the input before relying on the total.*

When a question gives a complete workflow, mark the stage of each action rather than assigning one label to the entire system. Entering a value is input; checking or summing it is processing; showing a result is output; keeping the transaction record is storage. If the stored record is opened and processed again, its role changes to input for that later operation.

Exercises

1. A user enters four prices, applies a tax formula, prints the invoice, and saves the file. Identify the input, processing, output, and storage.
2. Explain why a fast, consistent computer can still produce an incorrect result.
3. Three temperature readings are 18, 20, and 22 degrees. For a task asking for their average, identify the data and the resulting information.
4. Give one example from banking or governance and trace it through IPOS.

Solutions

1. The four prices are input; applying the tax formula is processing; the printed invoice is output; saving the invoice file is storage.
2. The computer follows its instructions and operates on the supplied data. Incorrect input or a wrong formula can therefore produce an incorrect result consistently; speed does not establish truth.
3. The three readings are input data. Their average is $(18 + 20 + 22)/3 = 20$ degrees and is information for the average-reading task.
4. Example: an online fee form accepts the user's details and amount (input), calculates the fee (processing), shows a receipt (output), and saves the transaction record (storage).